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RESEARCH-ARTICLE

IntoTheRabbitHole: A Web Application for Teaching Middle School Students About Search Algorithms

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IntoTheRabbitHole: A Web Application for Teaching Middle School Students About Search Algorithms

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Abstract

As artificial intelligence (AI) technology rapidly advances and integrates further into everyday life, building AI literacy in children is now more important than ever. AI literacy includes understanding key AI concepts such as search algorithms, which are foundational and have wide-ranging real-world applications. To support curricula that foster AI and computational literacy in K-12 students, we developed a novel, easy-to-use software tool to introduce this foundational topic by teaching two search algorithms to middle school children. We present IntoTheRabbitHole, an interactive web application where users help a rabbit find its carrot to learn about Depth First Search (DFS) and Breadth First Search (BFS). In this work, our research questions were: (RQ1) Can we develop an engaging and enjoyable web application to teach search algorithms?; (RQ2) Could we find evidence of student learning of Depth First Search and Breadth First Search from using our application?; and (RQ3) How do students compare Depth First Search and Breadth First Search after interacting with IntoTheRabbitHole? We collected both quantitative and qualitative data on student use of IntoTheRabbitHole at an after-school AI program. We found that IntoTheRabbitHole was successful in fostering an understanding of both DFS and BFS. Furthermore, we obtained a statistically significant result showing that interaction with IntoTheRabbitHole led to better performance on a DFS traversal pre- and post-survey item, indicating that many students learned DFS through interaction with our tool.

CCS Concepts

• **Social and professional topics** → **K-12 education**; • **Computing methodologies** → **Knowledge representation and reasoning**; • **Applied computing** → **Interactive learning environments**.

Keywords

Search Algorithms, AI Literacy, K-12 AI Education, Artificial Intelligence

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1 Introduction

As artificial intelligence (AI) becomes increasingly widespread in everyday life [6, 26], children are now growing up surrounded by AI technologies, including social media filters, large language models, recommendation systems, and more [11]. To ensure children become responsible users of these applications [5, 14] and are prepared for a future immersed in even more AI technology [20, 25], it is important to introduce AI concepts at a young age to develop their AI literacy and computational literacy early-on [4, 8, 12, 23, 27]. This is because early-life exposure can help children better understand the inner workings and limitations of AI [7]. To that end, we developed IntoTheRabbitHole, an interactive web application to introduce middle schoolers to two foundational search algorithms: Depth First Search (DFS) and Breadth First Search (BFS). In this work, our research questions were: (RQ1) Can we develop an engaging and enjoyable web application to teach search algorithms?; (RQ2) Could we find evidence of student learning of Depth First Search and Breadth First Search from using our application?; and (RQ3) How do students compare Depth First Search and Breadth First Search after interacting with IntoTheRabbitHole? Our findings suggest that IntoTheRabbitHole can be used to introduce children to DFS/BFS, and we hope it will support initiatives to teach middle schoolers about AI in the classroom.

2 Related Work

As the significance of K-12 AI education rises, it becomes crucial to consider which topics to introduce to children [13]. Search algorithms are an essential topic to teach, as they are foundational and have wide-ranging real-world applications. Their significance is emphasized by their inclusion in cornerstone educational resources, like the *Artificial Intelligence: A Modern Approach* textbook [17] and AI4K12.org’s progression charts [1]. Search algorithms are also included in many other courses, like Sperling and Lickerman’s AI-infused high school software engineering course [18] and Kandhofer et al.’s AI modules for K-12 students [7].

Touretzky et al. outlined five “Big Ideas in AI” to teach K-12 students, with search algorithms falling under Big Idea 2 (“Agents maintain models/representations of the world and use them for reasoning”) [22]. AI4K12.org created progression charts for each “Big Idea,” detailing what students at different grade levels should learn. In Big Idea 2’s chart [1], different search concepts are included, emphasizing that K-12 students should learn them, but at age-appropriate levels of detail. It highlights that students should be

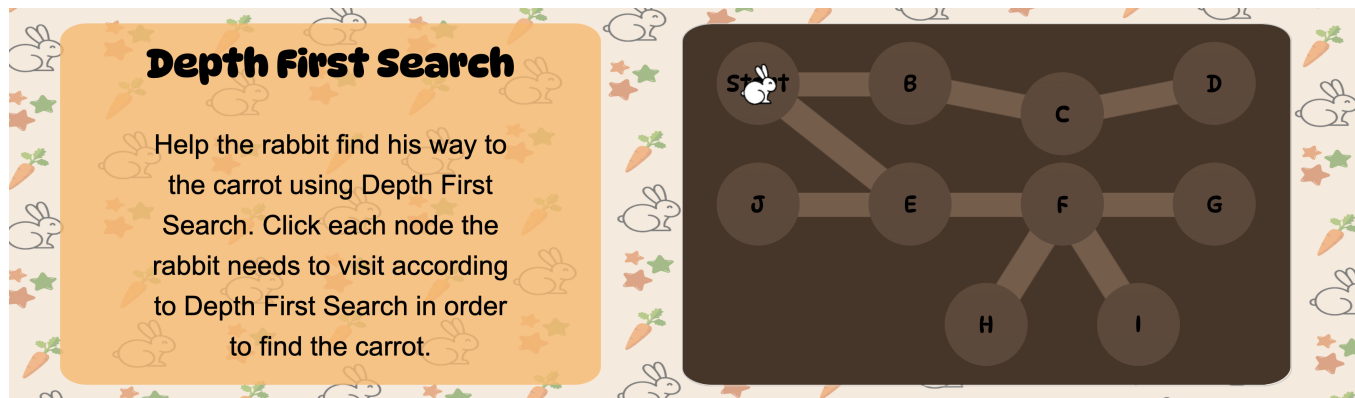


Figure 1: User interface of a DFS Puzzle Training Level

able to: draw potential states in tic-tac-toe in grades K-2, define tic-tac-toe states in grades 3-5, create detailed search spaces to navigate mazes in grades 6-8, and identify real-world search problems and discuss their states in grades 9-12. Big Idea 2's chart further specifies the search algorithm to be taught at a given grade-level: BFS for grades 6-8, and DFS, BFS, and best-first search for grades 9-12. The inclusion of search representations and algorithms in Big Idea 2's chart demonstrates the significance of search and the capability students have to grasp various aspects of search problems.

Goslen et al. developed and tested three activities to teach middle schoolers various AI concepts in a summer program [3]. One activity taught DFS, BFS, and A* search by visualizing them on mazes. Rather than having children directly interact with search algorithms, they focused on using visualizations. The study found that while students understood A* was the most efficient, they were unable to give detailed comparisons of the three algorithms.

Akram et al. created an AI curriculum for middle schoolers, including activities to teach BFS such as: introduction to graph traversal, BFS visualization/explanation, BFS traversal exercises, filling in missing components of BFS block-based code, and an activity connecting BFS to a real-world application [2]. Findings showed that students successfully completed even the code activity.

In the AI4GA project, Touretzky et al. are piloting an AI elective for middle schoolers [21]. They included instruction on BFS using unplugged activities, such as a graph-coloring-based route-finding task and introduction to search trees. Their work showed that middle schoolers can grasp technical concepts, like BFS.

Leitner et al. developed ARIN-561, an interactive game for high schoolers where they use search algorithms to help a character complete tasks and return to their home planet [10]. ARIN-561 was tested in multiple studies and showed strong results in improving student comprehension of DFS, BFS, and greedy search [9, 24].

Multiple approaches to teaching search algorithms have successfully fostered an understanding of DFS/BFS in K-12 students. Goslen et al. demonstrated the value of visualizations in introducing search algorithms to children, while also showing its limitations in fostering a deeper understanding [3]. Akram et al. highlighted that middle schoolers can develop a strong understanding of BFS when provided with thorough educational resources [2]. Touretzky et al. and Leitner et al. showcased the effectiveness of engaging, activity-

or game-based approaches to teaching children AI concepts, including search algorithms [9, 21].

While Goslen et al.'s work focused on using visualizations, our approach is more interactive, by also incorporating approaches highlighted by Touretzky et al. and Leitner et al. We developed an easily accessible and easy-to-use web application for teaching search algorithms that can be used in contexts beyond this study. Additionally, our application collects quantitative application interaction data to assess middle schoolers' learning of DFS/BFS, addressing a gap in prior approaches that focus primarily on observational or qualitative survey data. Furthermore, IntoTheRabbitHole was designed as a flexible introductory tool for DFS/BFS that can be supplemented with additional material in educational contexts outside this study.

3 Tool Design

IntoTheRabbitHole is an interactive web application for introducing middle schoolers to two search algorithms, DFS and BFS, by using them to help a rabbit find its carrot. To make it easy-to-use, IntoTheRabbitHole was developed as a web-based tool, using HTML, CSS, and Javascript. Source code is available at [15] and live version is deployed at [16]. For research purposes, IntoTheRabbitHole collects quantitative user interaction data (described in section 4.1).

3.1 Graph Traversal Puzzles

In IntoTheRabbitHole, each puzzle involves traversing the rabbit's tunnel system (a graph) to find a carrot. The carrot's location is hidden, simulating how the location of the goal state is initially unknown to AI agents. Once the user reaches the correct node, the carrot is revealed, and a "good job" message is displayed.

Each puzzle specifies one search algorithm and randomly selects a node to hide the carrot in. Users traverse the graph by clicking on nodes the rabbit must visit as it searches for the carrot. The rabbit only moves to a clicked node if it was correct according to that puzzle's search algorithm, otherwise a "try again" message appears.

3.1.1 Distinction Between Traversal with Canonical DFS/BFS and IntoTheRabbitHole DFS/BFS. Canonical BFS involves expanding the initial node, then its child nodes, then those nodes' child nodes, and so on. Expanded nodes are stored in a frontier (first-in-first-out

Table 1: Canonical versus IntoTheRabbitHole DFS/BFS traversal of the graph in Figure 1

	Canonical Traversal	IntoTheRabbitHole Traversal
DFS	Start, B, C, D, E, F, G, H, I, J	Start, B, C, D, C, B, Start, E, F, G, F, H, F, I, F, E, J
BFS	Start, B, E, C, F, J, D, G, H, I	Start, B, Start, E, Start, B, C, B, Start, E, F, E, J, E, Start, B, C, D, C, B, Start, E, F, G, F, H, F, I

queue) and searched in that order [17]. Canonical DFS, in contrast, uses a last-in-first-out queue as frontier so the deepest nodes are checked first. DFS and BFS both store explored nodes in a set, so the same node is not revisited [17].

IntoTheRabbitHole’s graphs represent real-world tunnels, and the rabbit (as agent) adheres to real-world logic. For instance, in IntoTheRabbitHole’s DFS/BFS traversal, the rabbit can only move between nodes directly connected by an edge. In contrast, canonical DFS/BFS allows the agent to move between non-adjacent nodes if they are next in the traversal path. Table 1 shows the difference between IntoTheRabbitHole and canonical DFS/BFS traversal of the graph in Figure 1. The backtracking the rabbit must follow in IntoTheRabbitHole adds complexity to DFS/BFS paths but was retained as a useful abstraction to introduce DFS/BFS at a high-level.

3.2 User Experience

First, the user sees a brief explanation of DFS and an animation of a rabbit performing DFS traversal on the **DFS Animation Page**. This animation loops so the user can watch it multiple times.

The next three pages are **DFS Puzzle Training Levels** (user interface shown in Figure 1). On each page, difficulty is equal, and the location of the hidden carrot is randomly selected. Having three pages ensures users complete at least three DFS puzzles. After the third DFS puzzle, users can choose to “Play Again” for more puzzles or proceed to the next level. Thus, each user completes at least three DFS puzzles here and some users do more.

Next is the **BFS Animation Page**, which shows a short explanation and traversal animation of BFS, similar to the DFS Animation Page. The user then encounters three **BFS Puzzle Training Levels**, similar to the DFS Puzzle Training Levels.

Last is the timed **Challenge Level**, where users complete as many DFS and BFS puzzles as possible in 60 seconds, while aiming for speed and accuracy. This level can be replayed multiple times.

4 Study Design

IntoTheRabbitHole was tested with middle schoolers (grades 6–8) at a STEM public charter school in Texas, with IRB approval, parental consent, and student assent. It was presented at a two-day after-school AI Expo where nine projects, each developed to teach students about AI concepts, were showcased. Students interacted and rotated between projects, including IntoTheRabbitHole. At the start of each session, students were given a pre-survey, after which they rotated between projects. Due to time constraints, not all students interacted with every project. Out of 109 participants, 49 interacted with IntoTheRabbitHole.

For students who interacted with IntoTheRabbitHole, we provided a project overview and encouraged them to ask questions and have peer discussions as they played through. High-level descriptions of both DFS and BFS were also provided to aid student learning. Finally, students were given a post-survey. Most students spent ~8 minutes with our project, including ~5 minutes for application interaction and ~3 minutes for post-survey.

4.1 Data Collection

Quantitative and qualitative data were collected to assess learning.

A collective **pre-survey** (not specific to IntoTheRabbitHole) was administered to all AI Expo participants. It measured general attitudes towards AI and familiarity with AI concepts. One item was a DFS graph traversal problem to assess familiarity with DFS. Due to time constraints, a BFS item was not included in the pre-survey.

During participant interactions with IntoTheRabbitHole, our web application collected **interaction data** for each puzzle completed by a user, including the level, time spent, and number of correct and incorrect clicks. After the expo, this data was processed to collect additional aggregate data for each user, such as average accuracy across DFS puzzles (discussed in section 5.2.1).

Audio and screen activity, from the laptops students used to play IntoTheRabbitHole, were also recorded to collect discussions between participants or between the participants and researchers.

We also collected data through a **post-survey**. Unlike the pre-survey, this post-survey was specific to IntoTheRabbitHole. Post-survey items assessed both student enjoyment of the application and understanding of DFS/BFS. The DFS traversal problem from the pre-survey was also included.

5 Results

After reviewing the aforementioned data, we found the post-survey, IntoTheRabbitHole website interaction, and audio/screen activity data to be the most useful in answering our research questions.

5.1 (RQ1) Can we develop an engaging and enjoyable web application to teach search algorithms?

To assess participants’ enjoyment of IntoTheRabbitHole, we analyzed responses to a post-survey item asking students to rate how much fun they had from 1 (least fun) to 5 (most fun). As per Table 2, most students gave IntoTheRabbitHole a rating of at least 3, with a mean rating of 3.85. Additionally, 28.6% of students gave IntoTheRabbitHole a rating of 5, indicating they had a great deal of fun with it. We also observed many participants express their enjoyment of IntoTheRabbitHole, in ways such as below:

- Many students enjoyed the game-like nature of the timed challenge level and competed to get the highest score.
- One student especially loved the timed challenge level, playing it repeatedly to beat the high score. He accurately explained DFS and BFS to friends and exclaimed “I loved this!”
- When asked by the researchers, some students engaged in lively discussions on whether DFS or BFS would be more helpful for a lost person.

Table 2: Results of post-survey item “On a scale from 1 to 5, how fun was IntoTheRabbitHole (1 being least fun and 5 being most fun)?” (Mean = 3.85, Standard Deviation = 1.04)

Rating	No.
1	2
2	2
3	11
4	18
5	14
No Response	2

5.2 (RQ2) Could we find evidence of student learning of Depth First Search and Breadth First Search from using our application?

We analyzed both quantitative and qualitative data to measure student learning of DFS/BFS. Students spent an average of 4.82 minutes (standard deviation = 1.62 minutes) playing IntoTheRabbitHole.

5.2.1 DFS Learning Outcomes. To evaluate student learning of DFS, we examined their average accuracy scores across all DFS puzzles they completed. Each student’s average accuracy across DFS puzzles was calculated as:

$$\text{Average Accuracy}_{\text{DFS}} = \frac{\text{Total Number of Correct Clicks}_{\text{DFS}}}{\text{Total Number of Clicks}_{\text{DFS}}}$$

Each click a user makes to move the rabbit to a selected node is either correct or incorrect based on the traversal algorithm. The DFS average accuracy is calculated by considering all clicks across all DFS puzzles completed by the user. As per Figure 2, 18 of 49 students achieved average accuracy between 80-90%, and 20 students scored $\geq 90\%$, showcasing a strong understanding of DFS traversal.

Next, we analyzed responses to the DFS traversal item on both the pre- and post-survey that asked students to select the correct DFS traversal path for the graph shown. As per Table 4, pre-survey responses were similarly distributed among answer choices, with 28.6% of students answering correctly. On the post-survey, 67.3% of students answered this item correctly. To determine if there was a statistically significant difference in the proportion of students who answered this item correctly between the pre- and post-survey, we performed a McNemar’s Test. This test is used for paired binary data to assess whether there is a significant difference between two related groups [19]. Our data yielded a p-value of 0.001315, indicating that with 99% confidence, there was a statistically significant difference in the proportion of students who answered the DFS traversal item correctly on the pre-survey versus the post-survey. Therefore, interaction with IntoTheRabbitHole improved students’ performance on the DFS traversal item. Furthermore, we calculated the effect size, which is the ratio between number of responses that changed from incorrect-to-correct versus correct-to-incorrect. Specifically, 23 responses shifted from incorrect-to-correct, while 5 responses shifted from correct-to-incorrect, yielding an effect size of 4.6, which exceeds one. This demonstrates a greater likelihood that students change their answer from incorrect-to-correct after interacting with IntoTheRabbitHole, which is indicative of learning.

Table 3: Categorized responses to open-ended post-survey item that had students describe DFS in their own words.

Category	No.	Example Response
DFS Good Responses	13	“DFS is where you check in all possible spots in one direction, then backtrack and do the same thing over and over.”
Discuss purpose of DFS	9	“it finds pathways to find a goal”
Mention DFS is fun, cool, or helpful	7	“it is really cool”
Talk about direction	4	“first go horizontal and then go vertical”
Mention complicated	5	“complicated”
Alphabetical misconception	4	“you go in alphabetical order”
Trivial, I don’t know, off topic, or no answer	7	“idk”

Table 4: Responses to the pre- and post-survey DFS traversal item. The correct answer is in bold.

Answer	Pre-survey No.	Post-survey No.
Start, B, C, D, C, B, Start, E, F	14	33
Start, B, Start, E, Start, B, C, B, Start, E, F	10	6
Start, E, J, E, F	17	1
Start, B, C, D, E, F	5	6
No response	3	3

In addition, we examined responses to an open-ended post-survey item that asked students to describe DFS. As per Table 3, we found 7 major categories of responses. Overall, 26.5% of responses explained DFS functionality well, with examples below:

- “It chooses a random path, then travels through that one until it hits a dead end or finishes. Then it travels through another path and continues the process until it has found the right path.”
- “DFS is where you check in all possible spots in one direction, then backtrack and do the same thing over and over.”
- “You go down each path and back through trial and error. If you hit a dead end, you go back to the last diversion in the path and go down that, and so on until you find whatever it is you are searching for.”

In addition, as per Table 3, 9 students understood the purpose of DFS and 7 students mentioned it was fun and helpful.

5.2.2 BFS Learning Outcomes. To analyze student learning of BFS, we first examined their average accuracies across all BFS puzzles. Like DFS average accuracy (discussed earlier), the BFS average accuracy for each student was calculated as:

$$\text{Average Accuracy}_{\text{BFS}} = \frac{\text{Total Number of Correct Clicks}_{\text{BFS}}}{\text{Total Number of Clicks}_{\text{BFS}}}$$

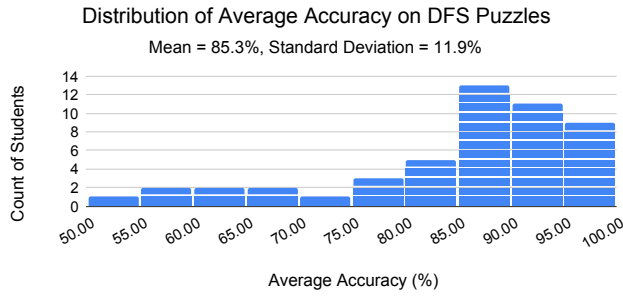


Figure 2: Distribution of per-student DFS average accuracies.

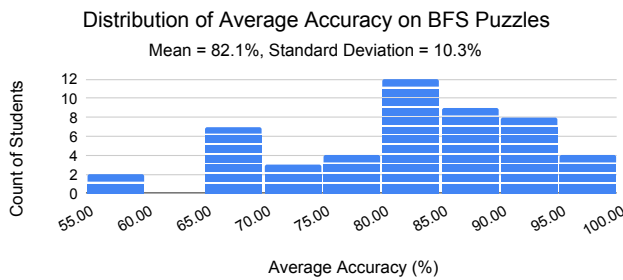


Figure 3: Distribution of per-student BFS average accuracies.

Table 5: Responses to the post-survey BFS traversal item. The correct answer is in bold.

Answer	No.
Start, B, Start, E, Start, B, C, B, Start, E, F	26
Start, B, Start, E, Start, B, C, D, C, B, Start, E, F	8
Start, B, Start, E, F	6
Start, B, C, D, C, B, Start, E, F	7
No response	2

As seen in Figure 3, 21 of 49 students achieved an average accuracy between 80-90%, indicating a moderate understanding of BFS traversal. Additionally, 12 students scored an average accuracy $\geq 90\%$, indicating a strong understanding of BFS traversal.

Next, we analyzed responses to a post-survey item that displayed a graph and asked students to select the correct BFS traversal path. As seen in Table 5, 53.1% of students answered correctly.

We then examined responses to an open-ended post-survey item that asked students to describe BFS. Of these, 17.4% provided thorough descriptions of BFS functionality, with examples below:

- “Breadth First Search is where you check the spots closest to you and slowly work your way farther and farther.”
- “It is an artificial [intelligence] method covering each obstacle from closest to the starting point, coming back each time and slowly working its way to achieve the goal.”
- “The AI starts based on alphabetical order and works on all paths at the same time.”

Table 6: Responses to the post-survey multiple-choice item comparing the performance of DFS versus BFS and sample responses explaining their answer.

Response to Post-Survey Multiple Choice Item Comparing DFS and BFS	No.	Sample Open-Ended Response Explaining Their Selection
It depends on the graph/tunnel system and the location of the goal	21	“If the path is small, DFS is faster due to it being able to check every corner in the map at a decently fast pace, but if the maze is big, BFS is faster due to the processes ability to multi-task.”
They are both equally fast	7	“They both complete an equal amount of ground”
DFS is always faster	14	“DFS focuses on one area at a time, meanwhile, BFS kept backtracking.”
BFS is always faster	4	“I believe BFS is faster because you get to check every path efficiently”
No Response	3	

5.2.3 Student Learning of DFS versus BFS. Furthermore, we analyzed the data to compare student learning of DFS versus BFS. As shown in Figures 2 and 3, the average accuracy distribution for DFS had a higher mean (85.3%) than BFS (82.1%). Also, 40.8% of students achieved an average accuracy $\geq 90\%$ on DFS puzzles compared to only 24.5% achieving $\geq 90\%$ on BFS puzzles. In addition, the open-ended item asking students to describe DFS had a higher proportion of good responses (26.5%) compared to the BFS counterpart (17.4%).

5.3 (RQ3) How do students compare Depth First Search and Breadth First Search after interacting with IntoTheRabbitHole?

To understand how students compare DFS and BFS after using IntoTheRabbitHole, we analyzed responses to a post-survey multiple-choice item comparing DFS and BFS performance and accompanying open-ended explanations (shown in Table 6). The item had no single correct answer and was meant to gauge student perception of DFS versus BFS. The option “it depends on the graph/tunnel system and the location of the goal” recognizes that algorithm performance depends on the given graph, while “they are both equally fast” acknowledges that DFS and BFS have similar performances due to equal time complexity. Among the 49 students, 21 answered “it depends on the graph/tunnel system and the location of the goal” and 7 chose “they are both equally fast,” while 14 believed DFS was faster.

6 Discussion

With IntoTheRabbitHole, students learned about search algorithms by using them to help a rabbit find its carrot. Our work demonstrates that engaging and fun software tools can effectively introduce children to DFS/BFS, as most students enjoyed the IntoTheRabbitHole experience and gave it a mean rating of 3.85 out of 5.

Furthermore, we found evidence that most students understood DFS/BFS after interacting with IntoTheRabbitHole. Among 49 students, 18 demonstrated an average accuracy between 80–90% on DFS puzzles, indicating a solid DFS understanding, while 20 achieved average accuracy $\geq 90\%$, showing a strong understanding. In addition, using McNemar’s test, we found a statistically significant improvement in the proportion of students who answered the DFS traversal item correctly between the pre- and post-survey, with only five students changing their answer from correct-to-incorrect. Of these five, two likely confused DFS with BFS as they provided the BFS traversal path instead, and three may have struggled with the backtracking in IntoTheRabbitHole. Overall, there was a noticeable increase in the number of correct responses after interacting with IntoTheRabbitHole, showcasing its effectiveness in teaching DFS traversal. Also, 13 of 49 post-survey open-ended responses (26.5%) to the item asking students to describe DFS were thorough. Although IntoTheRabbitHole provided only a brief text explanation of DFS, these students’ open-ended responses demonstrated a strong understanding after interacting with IntoTheRabbitHole.

Next, 21 of 49 students achieved an average accuracy between 80–90% on BFS puzzles, indicating a moderate understanding of BFS. Notably, 12 students attained an average accuracy ≥ 90 , indicating a strong understanding of BFS traversal. Additionally, 17.4% of responses to the open-ended post-survey item asking students to describe BFS were evaluated as good quality. Given the brief text overview of BFS in IntoTheRabbitHole, these responses indicate after interacting with the application, some students developed a strong enough understanding of BFS to sufficiently explain it.

Student performance on DFS puzzles and survey items surpassed that of BFS. For example, 33 students (67.3%) correctly answered the DFS traversal item on the post-survey, compared to only 26 students (53.1%) for the BFS counterpart. As discussed in section 3.1.1, IntoTheRabbitHole’s DFS/BFS traversals are more complex than the canonical versions as the rabbit must walk from node to adjacent node. IntoTheRabbitHole’s BFS traversal path is even longer than our DFS path due to increased backtracking, making it harder for students to grasp BFS. Nonetheless, many students developed some understanding of BFS, though less than DFS.

Analyzing student perception of DFS and BFS revealed most students correctly recognized the algorithms’ performances are either similar or dependent on the specific graph the agent is searching. Among students, 21 thought it depends on the graph and 7 thought DFS/BFS are equally fast. Also, 14 students thought DFS is always faster, likely because backtracking makes IntoTheRabbitHole’s BFS traversal longer and more complex, making DFS appear faster.

Overall, our findings indicate that IntoTheRabbitHole can foster a solid foundation in DFS/BFS concepts within a short timeframe. Going beyond Goslen et al.’s work [3], we incorporated interactive activities alongside visualizations in our approach, finding that this enabled students to develop a deeper understanding of DFS/BFS. For instance, 26.5% of DFS descriptions and 17.4% of BFS descriptions provided by students were of good quality and demonstrated their ability to articulate DFS/BFS well. Additionally, student enjoyment and learning metrics from our interactive graph traversal puzzles support the findings of Touretzky et al. and Leitner et al., highlighting the effectiveness of activity- and game-based methods in introducing children to search algorithms [9, 21].

Furthermore, many students demonstrated signs of learning, even though the mean time spent interacting with IntoTheRabbitHole was only 4.82 minutes. This indicates IntoTheRabbitHole can effectively serve as a quick introduction to DFS/BFS. Additionally, IntoTheRabbitHole is a web-based tool that requires no complex installation process and is accessible via web browsers using a link, making it convenient for educators to incorporate into middle school class lessons. IntoTheRabbitHole can also be paired with additional educational materials, as discussed by Akram et al. [2], to develop a deeper understanding of search algorithms.

7 Limitations

One limitation of IntoTheRabbitHole is that canonical DFS/BFS are not shown. However, the project provides a foundation that can be built upon in future lessons. Additionally, although the carrot’s location is randomized, all graph traversal puzzles display the same graph. This was a deliberate choice because students had limited time to engage with IntoTheRabbitHole at the program. Hence, using multiple graphs may have confused students, so we prioritized helping them first achieve a high-level understanding of DFS/BFS. While this approach serves as an introduction to DFS/BFS, adding more graphs could better evaluate students’ ability to apply DFS/BFS and instill a deeper understanding. Another limitation is the nearly alphabetical lettering of nodes in the DFS traversal path that led some students to believe DFS searches alphabetically.

8 Conclusion and Future Work

In this study, we developed IntoTheRabbitHole, an interactive web application for teaching middle schoolers about two search algorithms: Depth First Search and Breadth First Search. The application was tested with middle schoolers in an after-school program. Our findings indicate that students developed a good understanding of DFS and a moderate understanding of BFS. Most students achieved average accuracies $\geq 80\%$ for both DFS and BFS puzzles, demonstrating a solid ability to use these search algorithms effectively. Additionally, using McNemar’s Test, we found a statistically significant improvement in the proportion of students correctly answering a DFS traversal item between the pre- and post-survey, demonstrating that interacting with IntoTheRabbitHole improved their DFS traversal performance. The effect size of 4.6 also showed a higher likelihood of students changing their DFS traversal item answers from incorrect-to-correct, showcasing that IntoTheRabbitHole was successful in effectively teaching many students DFS.

Future studies could address limitations by testing with more participants, introducing canonical DFS/BFS, incorporating more complex graphs, and randomizing node lettering in graph traversal puzzles. In this study, a BFS item was not included in the pre-survey due to time constraints. Incorporating this in future work will enable pre- and post-survey analysis of BFS, providing deeper insights into student learning of BFS.

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